| INCH-POUND |

MIL-C-55365/2D
23 August 1990
SUPERSEDING
MIL-C-55365/2C
10 March 1987

MILITARY SPECIFICATION SHEET

CAPACITORS, CHIP, FIXED, TANTALUM, ESTABLISHED RELIABILITY, STYLES CWR03 AND CWR04

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.

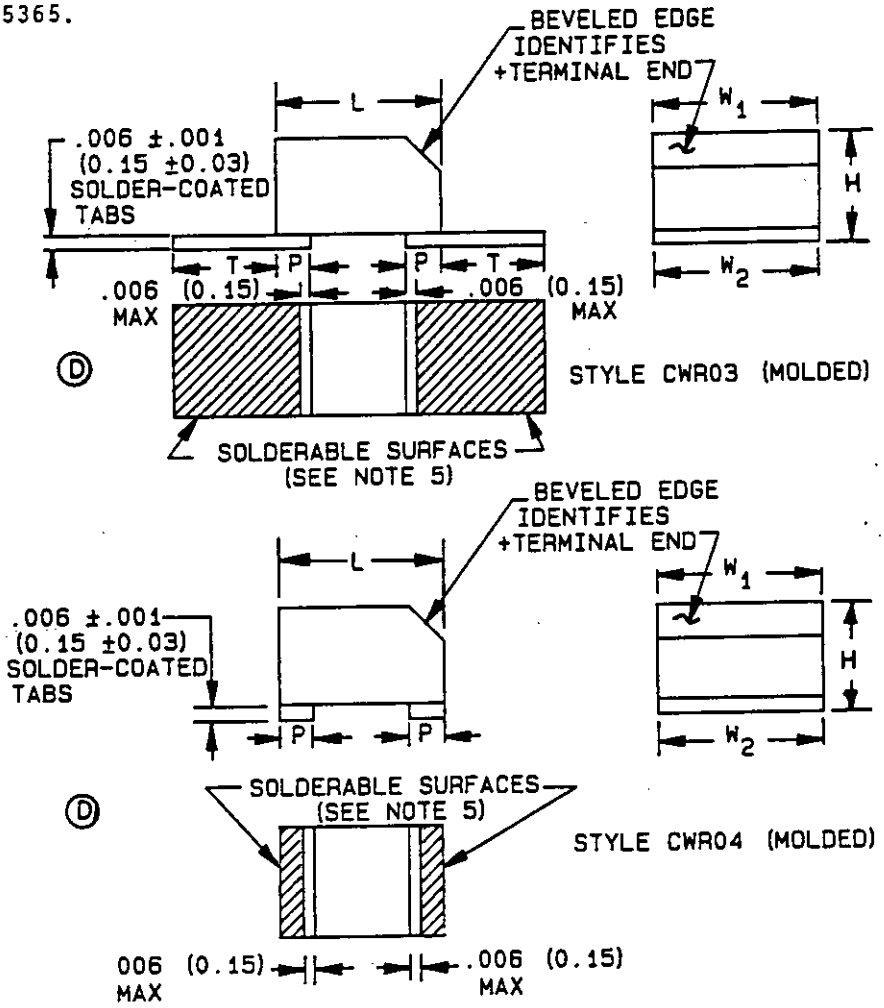


FIGURE 1. Styles CWR03 and CRW04 capacitors.

(D) denotes changes

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Case size	L	(Case) W_1	(Tabs) W_2	H	P	T
	±.005 (0.13)	±.010 (0.25)	±.010 (0.25) -.005 (0.13)	±.005 (0.13)	±.010 (0.25)	±.010 (0.25)
A	.180 (4.57)	.100 (2.54)	.090 (2.29)	.070 (1.78)	.030 (0.76)	.150 (3.81)
B	.180 (4.57)	.100 (2.54)	.090 (2.29)	.100 (2.54)	.030 (0.76)	.150 (3.81)
C	.320 (8.13)	.180 (4.57)	.170 (4.32)	.070 (1.78)	.050 (1.27)	.110 (2.79)
D	.320 (8.13)	.180 (4.57)	.170 (4.32)	.100 (2.54)	.050 (1.27)	.110 (2.79)
E	.320 (8.13)	.180 (4.57)	.170 (4.32)	.195 (4.95)	.050 (1.27)	.110 (2.79)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. These capacitors are designed for mounting by dip soldering, welding, reflow soldering, or other conventional means.
(D) 5. Solderable surfaces are only those surfaces designated as such. Tab edges are not considered solderable.

FIGURE 1. Style CWR03 and CWR04 capacitors - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case type: Chip.

Material: Tantalum.

(D) Leads: Solder coated tab.

(D) Termination finish: In accordance with termination finish C of MIL-C-55365.

DC rated voltage: See table I. Above +85°C, voltage derating is required (see MIL-C-55365).

Operating temperature range: -55°C to +125°C.

(D) Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: ±5 percent (J), ±10 percent (K), or ±20 percent (M).

Dissipation factor (DF): See table I.

(D) Resistance to soldering heat: In accordance with MIL-C-55365.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Shall not exceed tolerance of table I value.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Shall not exceed ±10 percent of initial measurement.
	DF	-	See table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±10 percent of initial +25°C measurement.
	DF	-	See table I.
Step 5 (+125°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±15 percent of initial +25°C measurement.
	DF	-	See table I.
Step 6 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Shall not exceed ±5 percent of initial +25°C measurement.
	DF	-	See table I.

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Surge voltage: In accordance with MIL-C-55365.

DCL - See table I.
ΔCap. - Within ±5 percent of initial value.
DF - See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At 25°C: DCL - See table I.
ΔCap. - Shall not exceed ±5 percent of initial measured value.
DF - See table I.

At 85°C: DCL - See table I.

At 125°C: DCL - See table I.

10,000 hours:

At 25°C: DCL - 200 percent of initial requirement.
ΔCap. - Shall not exceed ±10 percent of initial measured value.
DF - See table I.

At 85°C: DCL - 200 percent of initial requirement.

At 125°C: DCL - 200 percent of initial requirement.

Ⓓ TABLE I. Styles CWR03 and CWR04 characteristics.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	μF	μA	μA	μA	%	%	%	
CWR0-BC185--	3	1.8	1.0	10	15	4	6	8	A
CWR0-BC225--	3	2.2	1.0	10	15	4	6	8	A
CWR0-BC395--	3	3.9	1.0	10	15	4	6	8	B
CWR0-BC475--	3	4.7	1.0	10	15	4	6	8	B
CWR0-BC126--	3	12.0	1.0	10	15	6	8	10	C
CWR0-BC156--	3	15.0	1.0	10	15	6	8	10	C
CWR0-BC396--	3	39.0	2.3	23	35	6	8	10	D
CWR0-BC476--	3	47.0	2.8	28	42	6	8	10	D
CWR0-BC826--	3	82.0	4.9	49	74	8	10	12	E
CWR0-BC107--	3	100.0	6.0	60	90	8	10	12	E
CWR0-DC125--	6	1.2	1.0	10	15	4	6	8	A
CWR0-DC155--	6	1.5	1.0	10	15	4	6	8	A
CWR0-DC275--	6	2.7	1.0	10	15	4	6	8	B
CWR0-DC335--	6	3.3	1.0	10	15	4	6	8	B
CWR0-DC825--	6	8.2	1.0	10	15	6	8	10	C
CWR0-DC106--	6	10.0	1.2	12	18	6	8	10	C
CWR0-DC276--	6	27.0	3.2	32	48	6	8	10	D
CWR0-DC336--	6	33.0	4.0	40	60	6	8	10	D
CWR0-DC566--	6	56.0	6.7	67	101	8	10	12	E
CWR0-DC686--	6	68.0	8.2	82	123	8	10	12	E
CWR0-FC824--	10	0.82	1.0	10	15	4	6	8	A

See footnote at end of table.

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(D) TABLE I. Styles CWR03 and CWR04 characteristics - Continued.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR0-FC105--	10	1.0	1.0	10	15	4	6	8	A
CWR0-FC155--	10	1.5	1.0	10	15	4	6	8	B
CWR0-FC185--	10	1.8	1.0	10	15	4	6	8	B
CWR0-FC225--	10	2.2	1.0	10	15	4	6	8	B
CWR0-FC565--	10	5.6	1.1	11	17	4	6	8	C
CWR0-FC685--	10	6.8	1.4	14	21	6	8	10	C
CWR0-FC126--	10	12.0	2.4	24	36	6	8	10	D
CWR0-FC156--	10	15.0	3.0	30	45	6	8	10	D
CWR0-FC186--	10	18.0	3.6	36	54	6	8	10	D
CWR0-FC226--	10	22.0	4.4	44	66	6	8	10	D
CWR0-FC336--	10	33.0	6.6	66	99	6	8	10	E
CWR0-FC396--	10	39.0	7.8	78	117	6	8	10	E
CWR0-FC476--	10	47.0	9.4	94	141	6	8	10	E
CWR0-HC564--	15	0.56	1.0	10	15	4	6	8	A
CWR0-HC684--	15	0.68	1.0	10	15	4	6	8	A
CWR0-HC125--	15	1.2	1.0	10	15	4	6	8	B
CWR0-HC395--	15	3.9	1.2	12	18	4	6	8	C
CWR0-HC475--	15	4.7	1.4	14	21	4	6	8	C
CWR0-HC825--	15	8.2	2.5	25	38	6	8	10	D
CWR0-HC106--	15	10.0	3.0	30	45	6	8	10	D
CWR0-HC276--	15	27.0	8.1	81	122	6	8	10	E
CWR0-JC274--	20	0.27	1.0	10	15	4	6	8	A
CWR0-JC334--	20	0.33	1.0	10	15	4	6	8	A
CWR0-JC394--	20	0.39	1.0	10	15	4	6	8	A
CWR0-JC474--	20	0.47	1.0	10	15	4	6	8	A
CWR0-JC564--	20	0.56	1.0	10	15	4	6	8	B
CWR0-JC684--	20	0.68	1.0	10	15	4	6	8	B
CWR0-JC824--	20	0.82	1.0	10	15	4	6	8	B
CWR0-JC105--	20	1.0	1.0	10	15	4	6	8	B
CWR0-JC185--	20	1.8	1.0	10	15	4	6	8	C
CWR0-JC225--	20	2.2	1.0	10	15	4	6	8	C
CWR0-JC275--	20	2.7	1.1	11	17	4	6	8	C
CWR0-JC335--	20	3.3	1.3	13	20	4	6	8	C
CWR0-JC475--	20	4.7	1.9	19	29	4	6	8	D
CWR0-JC565--	20	5.6	2.2	22	33	4	6	8	D
CWR0-JC685--	20	6.8	2.7	27	41	6	8	10	D
CWR0-JC126--	20	12.0	4.8	48	72	6	8	10	E
CWR0-JC156--	20	15.0	6.0	60	90	6	8	10	E
CWR0-JC186--	20	18.0	7.2	72	108	6	8	10	E
CWR0-JC226--	20	22.0	8.8	88	132	6	8	10	E
CWR0-MC104--	35	0.10	1.0	10	15	4	6	8	A
CWR0-MC124--	35	0.12	1.0	10	15	4	6	8	A
CWR0-MC154--	35	0.15	1.0	10	15	4	6	8	A
CWR0-MC184--	35	0.18	1.0	10	15	4	6	8	A
CWR0-MC224--	35	0.22	1.0	10	15	4	6	8	A
CWR0-MC274--	35	0.27	1.0	10	15	4	6	8	B
CWR0-MC334--	35	0.33	1.0	10	15	4	6	8	B
CWR0-MC394--	35	0.39	1.0	10	15	4	6	8	B

See footnote at end of table.

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(D) TABLE I. Styles CWR03 and CWR04 characteristics - Continued.

PIN 1/	DC rated voltage (85°C)	Cap. (nom)	DC leakage (max)			Dissipation factor (max)			Case size
			+25°C	+85°C	+125°C	+25°C	+85°C +125°C	-55°C	
	Volts	µF	µA	µA	µA	%	%	%	
CWR0-MC474--	35	0.47	1.0	10	15	4	6	8	B
CWR0-MC824--	35	0.82	1.0	10	15	4	6	8	C
CWR0-MC105--	35	1.0	1.0	10	15	4	6	8	C
CWR0-MC125--	35	1.2	1.0	10	15	4	6	8	C
CWR0-MC155--	35	1.5	1.1	11	17	4	6	8	C
CWR0-MC185--	35	1.8	1.3	13	20	4	6	8	D
CWR0-MC225--	35	2.2	1.5	15	23	4	6	8	D
CWR0-MC275--	35	2.7	1.9	19	29	4	6	8	D
CWR0-MC335--	35	3.3	2.3	23	35	4	6	8	D
CWR0-MC395--	35	3.9	2.7	27	41	4	6	8	D
CWR0-MC475--	35	4.7	3.3	33	50	4	6	8	E
CWR0-MC565--	35	5.6	3.9	39	59	4	6	8	E
CWR0-MC685--	35	6.8	4.8	48	72	6	8	10	E
CWR0-MC825--	35	8.2	5.7	57	86	6	8	10	E
CWR0-MC106--	35	10.0	7.0	70	105	6	8	10	E
CWR0-NC683--	50	0.068	1.0	10	15	4	6	8	A
CWR0-NC823--	50	0.082	1.0	10	15	4	6	8	A
CWR0-NC104--	50	0.10	1.0	10	15	4	6	8	B
CWR0-NC124--	50	0.12	1.0	10	15	4	6	8	B
CWR0-NC154--	50	0.15	1.0	10	15	4	6	8	B
CWR0-NC184--	50	0.18	1.0	10	15	4	6	8	B
CWR0-NC224--	50	0.22	1.0	10	15	4	6	8	C
CWR0-NC274--	50	0.27	1.0	10	15	4	6	8	C
CWR0-NC334--	50	0.33	1.0	10	15	4	6	8	C
CWR0-NC394--	50	0.39	1.0	10	15	4	6	8	C
CWR0-NC474--	50	0.47	1.0	10	15	4	6	8	C
CWR0-NC564--	50	0.56	1.0	10	15	4	6	8	C
CWR0-NC684--	50	0.68	1.0	10	15	4	6	8	C
CWR0-NC824--	50	0.82	1.0	10	15	4	6	8	D
CWR0-NC105--	50	1.0	1.0	10	15	4	6	8	D
CWR0-NC125--	50	1.2	1.2	12	18	4	6	8	D
CWR0-NC155--	50	1.5	1.5	15	23	4	6	8	D
CWR0-NC185--	50	1.8	1.8	18	27	4	6	8	E
CWR0-NC225--	50	2.2	2.2	22	33	4	6	8	E
CWR0-NC275--	50	2.7	2.7	27	41	4	6	8	E
CWR0-NC335--	50	3.3	3.3	33	50	4	6	8	E

1/ Complete PIN shall include additional symbols to indicate style, capacitance tolerance, and FR level.

Marking: Capacitors shall be marked in accordance with method I of MIL-STD-1285. The capacitance, voltage, and "J" brand shall be indicated on the capacitors.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 17, 99
NASA - NA
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 11, 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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